

DSP VERSION 1.0

PAGE	INDEX
1	COVER PAGE
2	BLOCK DIAGRAM
3	VOLTAGE SOURCE
4	MCU

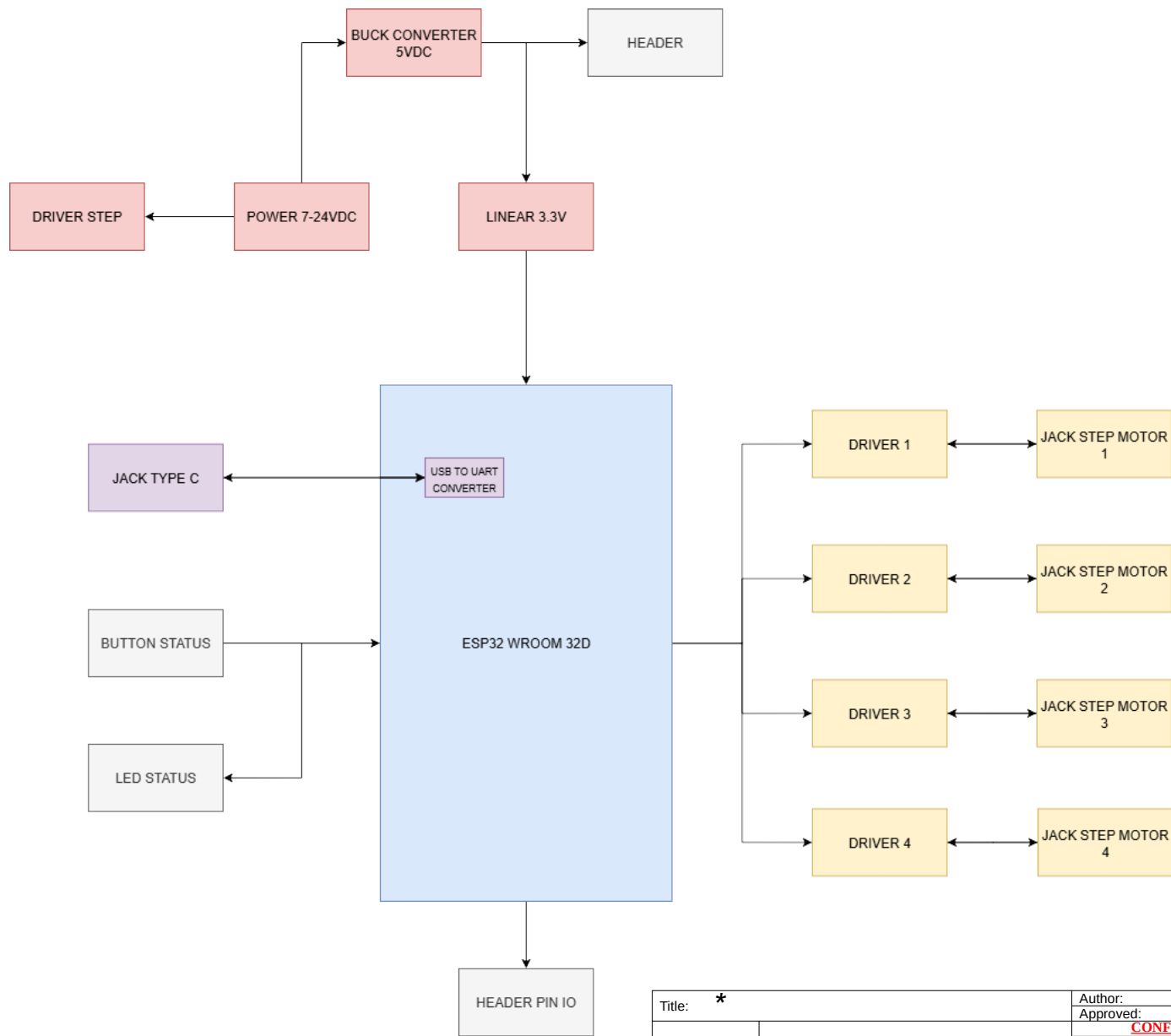
PAGE	INDEX
5	CONNECTOR
6	PERIPHERAL
7	
8	

DESIGN CONSIDERATIONS

DESIGN NOTE:
Functions of this block

DESIGN NOTE:
Calculation equations

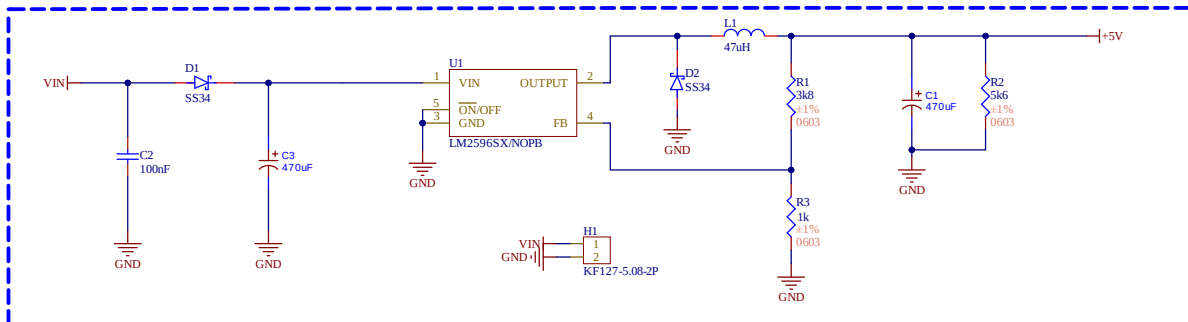
DESIGN NOTE:
Calculations when choosing
this component



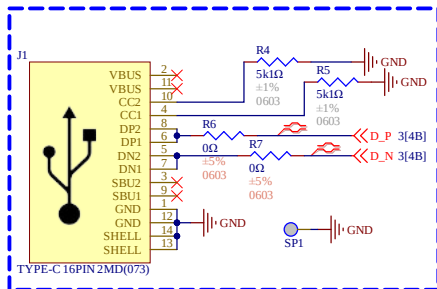
Title: *			Author: *
Size: A4			Approved: *
Project: DSP_VERSION_1.0			CONFIDENTIAL
Date: 4/7/2026 9:20:30 AM			Modified Date: 4/6/2026 *
Sheet 2 of 6			Revision: *
			SW Version: 24.1.2.44



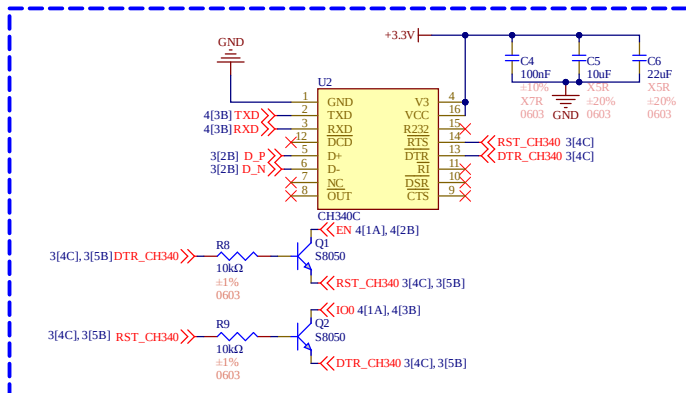
BUCK CONVERTER



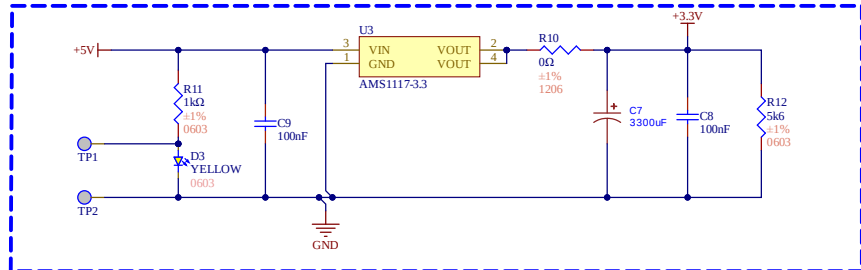
USB TYPE C



USB TO UART CONVERTER

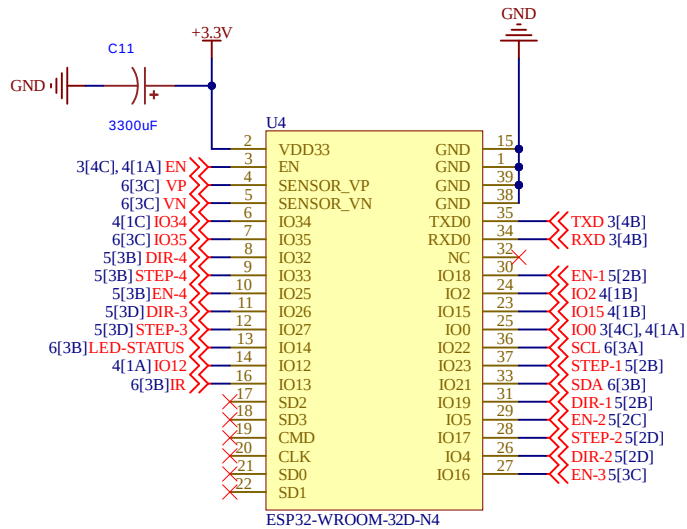
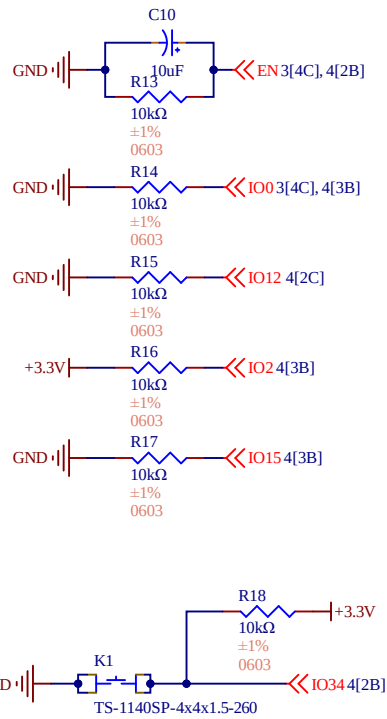


LINEAR MCU



Title: *		Author: *	
Size: A3	Project DSP_VERSION_10	CheckedBy: *	
Date: 4/7/2026	9:20:31 AM	Modified Date: 4/7/2026	
Sheet 3 of 6		Revision: *	
		SW Version: 24.1.2.44	

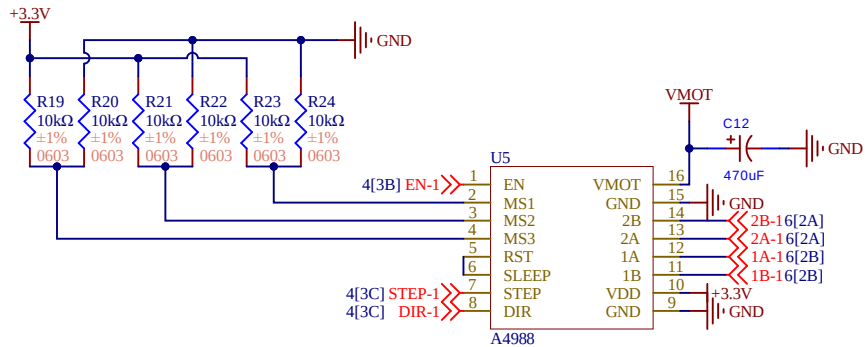




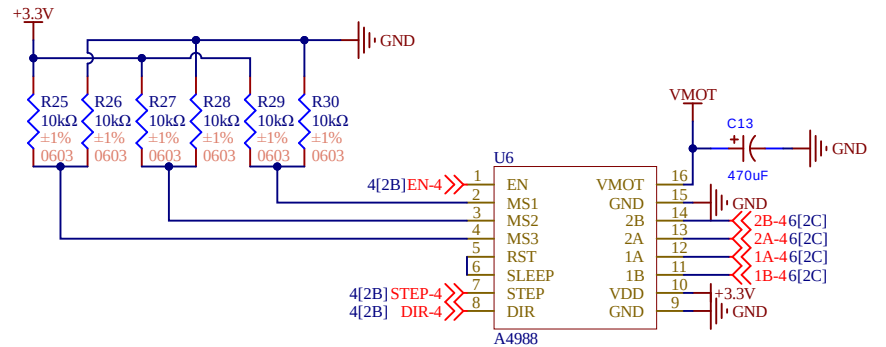
Title: ★		Author: *	
Size: A4		Approved: *	
Project: DSP_VERSION_1.0		CONFIDENTIAL	
Date: 4/7/2026		Modified Date: 4/7/2026	
9:20:31 AM		Revision: *	
Sheet 4 of 6		SW Version: 24.1.2.44	



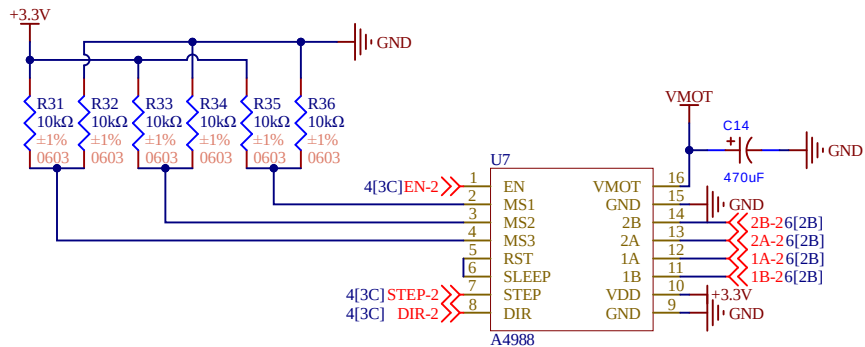
DRIVER STEP 1



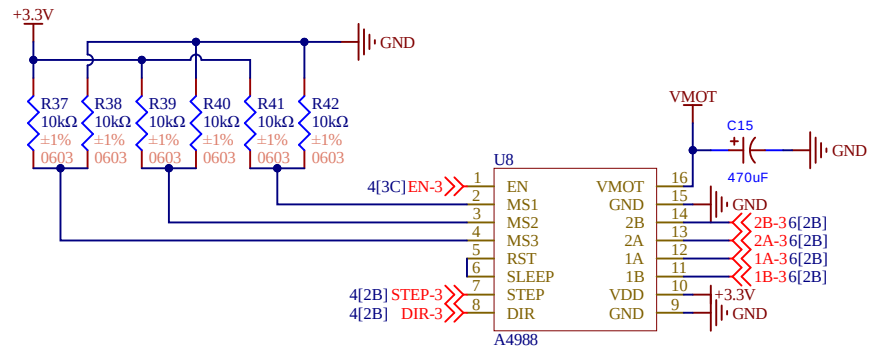
DRIVER STEP 4



DRIVER STEP 2



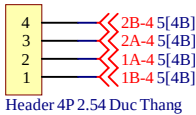
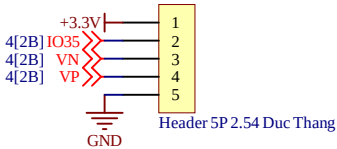
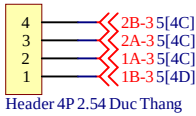
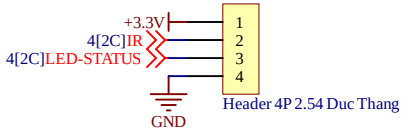
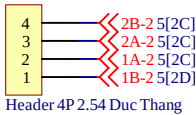
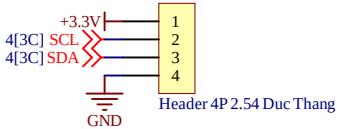
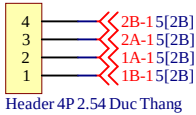
DRIVER STEP 3



Title: *			Author: *	
Size: A4			Approved: *	
Project: DSP_VERSION_1.0			CONFIDENTIAL	
Date: 4/7/2026			Modified Date: 4/7/2026	
9:20:32 AM			Revision: *	
Sheet 5 of 6			SW Version: 24.1.2.44	



	1	2	3	4
A				
B				
C				
D				
	1	2	3	4



Title: *			Author: *	
			Approved: *	
			CONFIDENTIAL	
Size: A4			Modified Date: 4/7/2026	
Project: DSP_VERSION_1.0			Revision: *	
Date: 4/7/2026	9:20:32 AM	Sheet 6 of 6	SW Version: 24.1.2.44	





Comment	Description	Designator	Footprint	Libref	Quantity
470uF	470uF ±20% 35V Through Hole.D10xL13mm	C1, C3, C12, C13, C14, C15	470uF 35V, D10xL13mm	470uF_35V, D10xL13mm	6
1000F	10m	C2, C8, C9	tu tantalum dip5.08	Tu tantalum 1000F 50V 20% DIP5.08	3
1000F	50V 100nF X7R ±10% 0603 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C4	C0603	C0603_100nF_±10%_X7R	1
10uF	25V 10uF XSR ±20% 0603 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C5	C0603	C0603_10uF_±10%_X7R	1
22uF	6.3V 22uF XSR ±20% 0603 Multilayer Ceramic Capacitors MLCC - SMD/SMT ROHS	C6	C0603	C0603_22uF_±20%_X5R	1
3300uF	3300uF ±20% 6.3V Through Hole.D8xL16mm 2000hrs	C7, C11	Through Hole.D8xL16mm	Through Hole.D8xL16mm	2
10uF	10uF 50V ±20% Through Hole.D5xL11mm Aluminum Electrolytic Capacitors - Leaded ROHS	C10	C_TH_5x11mm	C_10uF_5x11TH	1
SS34	40V independent 3A 550mV(@3A SMA/DO-214AC) Schottky Diodes ROHS	D1, D2	SS34	SS34	2
YELLOW	25mmx4mm Water Clear -40°C~+85°C 624nm Red 120° 60mW 0402 LED Indication - Discrete ROHS	D3	LED0603	D0603_YELLOW	1
KF127-5.08-2P	1.5 1x2P 14-26 Green 40°C~+105°C 15A 250V Through Hole 5.08mm 1.2 Through Hole-P=5.08mm Wire to Board Reels	H1	KF127-5.08-2P	KF127-5.08-2P	1
Header 4P 2.54 Duc Thang	10m	H2, H3, H4, H5, H6, H6	XH2.54M-4	Header 4P 2.54 Duc Thang	6
Header 5P 2.54 Duc Thang	10m	H7	XH2.54M-5	Header 5P 2.54 Duc Thang	1
TYPE-C 16PIN 2MD(073)	3A 1 Surface Mount, Right Angle 16P Female -25°C~+85°C Type-C SMD USB Connectors ROHS	J1	TYPE_C_FEMALE	TYPE_C_FEMALE_16PIN	1
TS-1140SP-4x4x1.5-260	SMD Tactile Switches ROHS	K1	4x4x1.5	TS-1140SP-4x4x1.5-260	1
47uH	1.15A 10uH Unshielded Inductor ±20% SMD,4x4.5mm Power Inductors ROHS	L1	CD43	CD4-10UH	1
S8050	25V 300mW 500mA SOT-23 Bipolar (BJT) ROHS	Q1, Q2	S8050	S8050	2
3k8	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R1	R0603	R0603_4k70_±1%	1
9k6	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R2, R12	R0603	R0603_4k70_±1%	2
1k	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R3	R0603	R0603_4k70_±1%	1
5k10	100mW Thick Film Resistor 75V ±1% ±100ppm/°C 100 5.1x0.8x0.3 Chip Resistor - Surface Mount ROHS	R4, R5	R0603	R0603_5k10_±1%	2
00	100mW Thick Film Resistor 75V ±5% 00 0603 Chip Resistor - Surface Mount ROHS	R6, R7	R0603	R0603_00_±5%	2
10k0	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R8, R9, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R29, R30, R31, R32, R33, R34, R35, R36, R37, R38, R39, R40, R41, R42	R0603	R0603_10k0_±1%	32
00	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R10	R1206	R1206_00_±1%	1
1k0	62.5mW Thick Film Resistor 50V ±1% ±200ppm/°C 100 0402 Chip Resistor - Surface Mount ROHS	R11	R0603	R0603_1k0_±1%	1
SP		SP1	SP	SP	1
TP		TP1, TP2	TP	TP	2
LM2596SX/NOPB	150kHz Buck Fixed 1 TO-263-5 Voltage Regulators - DC DC Switching Regulators ROHS	U1	LM2596SX-5.0/NOPB	LM2596SX/NOPB	1
CH340C	2Mbps Transceiver USB 2.0 SOP-16 USB Converters ROHS	U2	CH340C	CH340C	1
AMS1117-3.3	720mA(1.20Hz) 1A Fixed 3.3V Positive 16V SOT-223 Voltage Regulators - Linear, Low Drop Out (LDO) Regulators ROHS	U3	AMS1117-3.3	AMS1117-3.3	1
ESP32-WROOM-32D-N4	SMD,18x25.5mm WiFi Modules ROHS	U4	ESP32-WROOM-32D-N4	ESPRESSIF ESP32-WROOM-32D-N4	1
A4988		U5, U6, U7, U8	A4988-driver step	Module_A4988	4